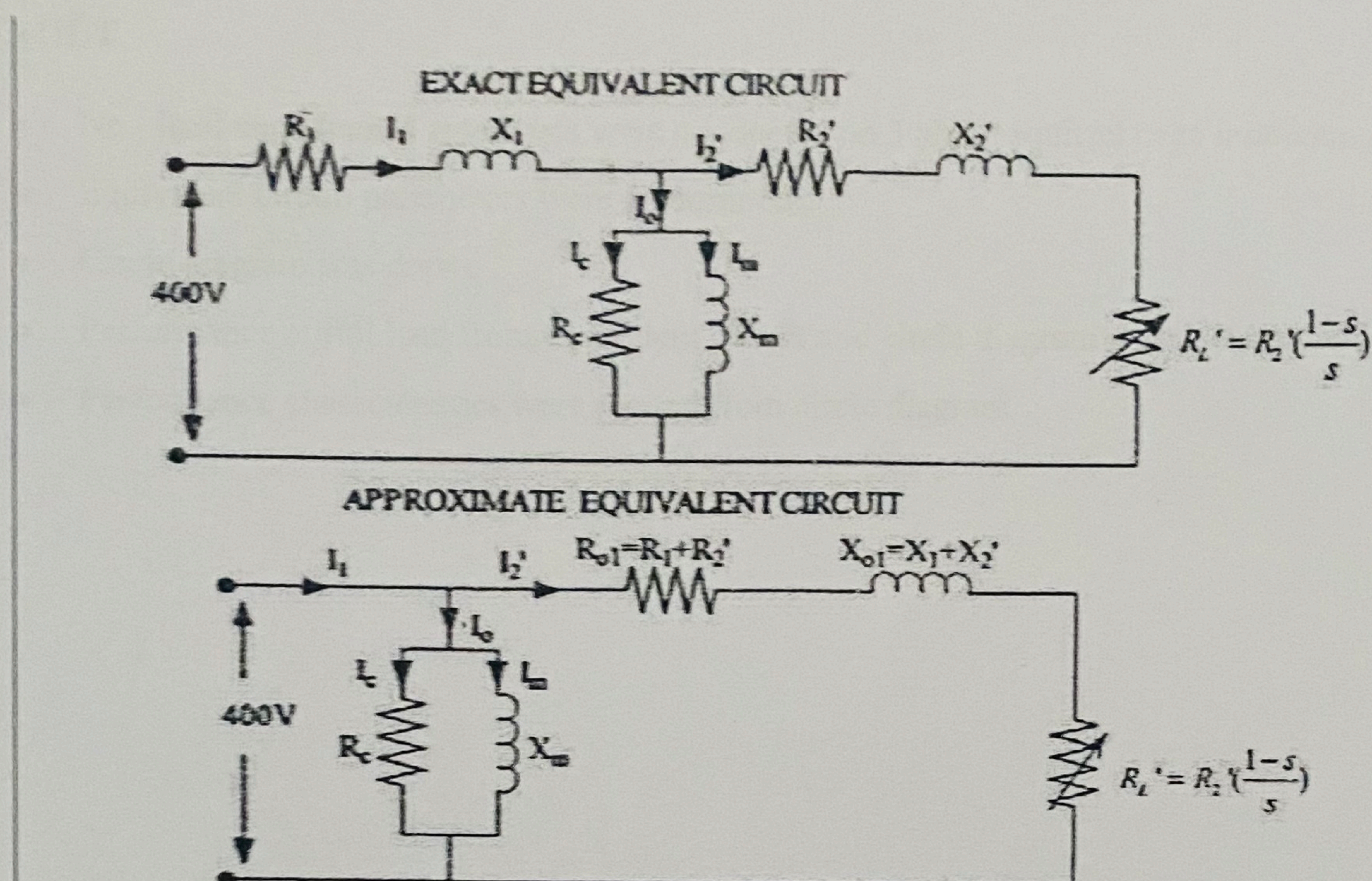




$$X_1 = X_2' = \frac{X_{01}}{2} = \underline{\hspace{2cm}} \Omega$$



PERFORMANCE AT RATED SPEED FROM EQUIVALENT CIRCUIT

Synchronous speed, $N_s =$ rpm

Rated speed = $N =$ rpm

$$\text{Slip, } s = \frac{N_s - N}{N_s} \times 100\% = \underline{\hspace{2cm}}\%$$

$$R_L' = R_2' \left(\frac{1-s}{s} \right) = \underline{\hspace{2cm}} \Omega$$

$$\bar{I}_0 = I_0 \angle (-\phi_0)^\circ = \underline{\hspace{2cm}} \text{ A per phase}$$

From approximate equivalent circuit,

$$\bar{I}_2' = \frac{V \angle 0^\circ}{(R_{01} + R_L') + jX_{01}} = \underline{\hspace{2cm}} \text{ A per phase}$$

$$\bar{I}_1 = \bar{I}_0 + \bar{I}_2' = I_1 \angle -\phi_1^\circ = \underline{\hspace{2cm}} \text{ A}$$

$$\text{Line current } I_L = \sqrt{3} I_1 = \underline{\hspace{2cm}} \text{ A}$$

$$\text{Power factor} = \cos(-\phi_1^\circ) = \underline{\hspace{2cm}}$$

$$\text{Output} = 3 \times I_2'^2 \times R_L' = \underline{\hspace{2cm}} \text{ W}$$

$$\text{Torque} = \frac{\text{output}}{(2\pi N/60)} = \underline{\hspace{2cm}} \text{ Nm}$$

$$\text{Input} = \sqrt{3} V I_L \cos \phi_1 = \underline{\hspace{2cm}} \text{ W}$$

$$\text{Efficiency} = \frac{\text{output}}{\text{input}} \times 100\% = \underline{\hspace{2cm}}\%$$